MILITARY SPECIFICATION

INSULATOR, BULKHEAD M753 8 INCH PROJECTILE

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for bulkhead insulators used in M753, 8 Inch Projectiles (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Deparment of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-843	Cushioning Material (Cellulosic)
PPP-T-42	Tape, Packaging/Masking, Paper

MILITARY

MIL-P-17667	Paper, Wrapping, Chemically Neutral (Non-Corrosive)
MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

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2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, publications form a part of this specification to the extent specified herein.

DRAWINGS (see 6.7)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

8806652	Marking and Sealing for Cylindrical Containers
9282999	Drum, Metal, Shipping (Exterior)
9328448	Insulator, Bulkhead
11829183	Molding Compound

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisiton functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D792	Specific Gravity and Density of Plastics by Displacement

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Construction. The bulkhead insulator shall comply with all requirements specified on drawing (dwg) 9328448 with all requirements specified herein and with the requirements of all applicable specifications to the extent specified on the dwg and herein.

3.2 Molding compound. Molding compound preforms shall comply with the requirements of dwg. 11829183, with the requirements of all applicable specifications and with the molding preform requirements herein. Preform molding

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compound shall be tested and approved by the Government (see 6.4) in accordance with the preform requirements herein prior to use in molding bulkhead insulators.

3.3 Molding preforms.

3.3.1 Chemical composition. The chemical composition of uncured preform molding compound shall be as indicated in Table I. Testing shall be performed in accordance with 4.5.1.1.

Table I. Chemical composition (preform)

Resin content	45 ± 3%
Fiberglass content	51 ± 4%
Volatile content	4 ± 1%

3.3.2 Infrared spectrum. The infrared spectrum of the resin extract of uncured preform molding compound shall qualitatively match the wavelength characteristics of Figure 1. Testing shall be performed in accordance with 4.5.1.2.

3.3.3 Curing properties. The curing properties of the preform molding compound shall be as specified in Table II (see Figure 2 for a typical plot of curing properties). Testing shall be performed in accordance with 4.5.1.3.

Table II. Curing properties (max).

Onset cure temperature (Tos)	125°C
Inflection point temperature (Tip)	133°C
Exothermic peak temperature (Texo)	140°C
Complete reaction temperature (Tcr)	151°C

3.4 Molded insulators.

3.4.1 Curing properties. Unless stored at or below 0°C, the curing properties of preform molding compound shall be retested and recertified to the requirements of 3.3.3 every 30 days.

3.4.2 Density. The density of the molded insulator shall be 1.70 ± .04 grams per cubic centimeter (gm/cc). Testing shall be performed in accordance with 4.5.2.1.

3.4.3 Transition temperature. The minimum transition temperature (Tg) of the molded insulator shall be 160°C. Testing shall be performed in accordance with 4.5.2.2.

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3.4.4 Flexure. With the periphery of the protruding side of the bulkhead insulator edge supported in an approved test fixture, there shall be no visible penetrating cracks or other physical defects when the center of the insulator is flexed a minimum of .025 inches at room ambient temperature. Minute surface cracks shall be acceptable provided they do not obviously penetrate below the glazed surface. Testing shall be performed in accordance with 4.5.2.3.

3.4.5 Voids and porosity. Molded insulators shall be free of internal voids or excess porosity when tested in accordance with 4.5.2.4.

3.4.6 Infrared inspection. Infrared inspection shall be performed on each bulkhead insulator in accordance with the requirements of dwg. 9328448 and 4.5.2.5.

3.5 First article. This specification contains technical provisions for first article inspection (see 4.3). Requirements for submission of first article samples shall be as specified in the contract.

3.6 Workmanship. Bulkhead insulators shall be uniform of finish and free of burrs and chipped edges. There shall be no evidence of obvious inclusions, blisters, cracks or other plastic defects likely to affect performance. Finished parts shall be clean and free of dirt, grease, abrasives or any other foreign matter. Cleaning methods shall not be injurious to the parts nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract or purchase order, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. When the contract requires first article inspection (see 3.5), a first article sample shall be submitted as designated by the contracing officer for evaluation in accordance with 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Molding Compound (Preforms)	11829183	3
Insulator, Bulkhead	9328448	10

4.3.2 Inspections to be performed. The first article sample submitted in accordance with 4.3.1 shall be subjected to any or all of the requirements of the applicable dwg. and to the tests and inspections specified in the First Article Inspection table herein. All parts shall have been manufactured using the same production processes, procedures, and equipment which shall be used in fulfilling the contract and all materials shall be obtained from the same source of supply as will be used in regular production.

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FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-I-63163(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.1	Insulator, Bulkhead				9328448
					NEXT HIGHER ASSEMBLY
					M753 Proj
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Classification of Defects and Tests.				
	Molding Compound Preforms	3			4.4.2.1
	Insulators, Bulkhead	10(a)			4.4.2.2
	NOTE: (a) After verification of density, dimensional characteristics and workmanship on all ten insulators, void and porosity tests shall be performed on five insulators, transition temperature tests shall be performed on one insulator and flexure tests shall be performed on the remaining four insulators. These are destructive tests.				

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.3.3 Inspection equipment. The inspections and tests specified in 4.3.2 shall be accomplished with the gages and test equipment specified in 4.4.4.

4.3.4 Rejection. See MIL-A-48078.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, an inspection lot of parts shall not exceed one days production and shall be produced from material and preforms from a single lot of molding compound.

4.4.2 Examinations. See MIL-A-48078 and the Quality Conformance Inspection Tables herein.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-I-63163(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Molding Compound Preforms (see note 1)				11829183
					NEXT HIGHER ASSEMBLY
					M753 Proj.
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101.	Chemical composition	(2)		3.3.1	4.5.1.1
102.	Infrared spectrum	(2)		3.3.2	4.5.1.2
103.	Curing properties	(2)		3.3.3	4.5.1.3

NOTES: (1) Government testing and approval required (see 3.2).
(2) One preform
(3) These are destructive tests.

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Insulator, Bulkhead	1 OF 2	9328448
			NEXT HIGHER ASSEMBLY: M753 Proj.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR /100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101.	Overall outside diameter		100%	3.1	Gage
102.	Overall height		100%	3.1	Gage
103.	Radius between flat face and overall outside diameter		100%	3.1	Gage
104.	Height of flange including step		100%	3.1	Gage
105.	Flatness of face		100%	3.1	Gage
106.	125 surface finish		100%	3.1	Visual
107.	Excessive shrinkage at referenced location		100%	3.1	Gage
108.	Outside diameter of flange step		-	3.1	Gage/TMC (note 1)
109.	Location of outside diameter of flange step		-	3.1	Gage/TMC
110.	Height of flange step		-	3.1	Gage/TMC
111.	Spherical protrusion (note 2)		-	3.1	Visual
112.	Outside diameter of protrusion at reference location (note 3)		-	3.1	Gage/TMC
113.	Location of protrusion		-	3.1	Gage/TMC
114.*	Curing properties (when required) (note 4)			3.4.1	4.5.1.3
115	Density		100%	3.4.2	4.5.2.1
116.*	Transition temperature	4.4 3.1		3.4.3	4.5.2.2

NOTES: (1) When TMC is specified, sample sizes shall be in accordance with 4.4.3.3.
(2) Visually verify that tip of protrusion is spherical (not flat) on TMC samples
(3) Verify at a minimum of two locations approximately 90 apart.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-I-63163(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2 (cont)	Insulator, Bulkhead	2 OF 2	9328448
			NEXT HIGHER ASSEMBLY: M753 Proj.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
117.*	Flexure	4.4 3.1	-	3.4.4	4.5.2.3
118.*	Voids and porosity	4.4 3.2		3.4.5	4.5.2.4
119.	Infrared inspection		100%	3.4.6	4.5.2.5
120.	Workmanship		100%	3.6	Visual

NOTES: (4) The curing properties of preforms shall be retested every 30 days. This retest shall not be required if molding preforms are stored at or below 0 C until used in molding.

(5) Tests preceded by an asterisk are destructive tests.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-I-63163(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Packaging				See 5.
					NEXT HIGHER ASSEMBLY
					N.A.
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101.	Packaging and packing not in accordance with applicable requirements		-	5.	Visual
102.	Objective evidence not available for shipping drum, packaging materials and adhesive tape		-	5.	Visual (see 6.5)
103.	Parts not clean and free of dirt and other foreign material prior to wrapping		100%	5.	Visual
104.	Parts not individually wrapped in wrapping paper and taped		100%	5.	Visual
105.	Parts not overwrapped in cushioning material and taped		100%	5.	Visual
106.	Shipping drum interior contaminated by rust, dirt or other foreign material		100%	5.	Visual
107.	Parts improperly packed in shipping drum		100%	5.	Visual
108.	Cushioning does not provide tight pack		100%	5.	Visual/Manual
109.	Shipping drum improperly sealed		100%	5.	Visual/Manual
110.	Marking missing, improper or illegible		100%	5.	Visual
111.	Poor workmanship		100%	-	Visual

NOTES:

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4.4.3 Testing. Testing is described in the first article and quality conformance inspection tables and herein.

4.4.3.1 Transition temperature, flexure. One sample insulator shall be randomly selected from each inspection lot of molded insulators (see lot definition in 4.4.1) and subjected to these tests in accordance with the applicable test method of 4.5. If any sample fails to meet the applicable requirements, the inspection lot of molded insulators represented by the sample shall be rejected. These are destructive tests and items may not be returned to the inspection lot.

4.4.3.2 Voids and porosity. For each mold cavity (after preheating), the first acceptable insulator from each days production shall be bisected and qualitatively examined for voids or excess porosity in accordance with 4.5.2.4. If the first insulator is found defective, corrective action shall be taken and each subsequently molded piece shall be examined in accordance with 4.5.2.4 until an acceptable part is produced before production of the inspection lot is initiated. These are destructive tests and items may not be returned to the inspection lot.

4.4.3.3 Tool manufacturing control. When tool manufacturing control (TMC) is specified, the first five pieces initially produced from each mold cavity inspected (after a minimum mold preheat of one hour) for the applicable classification of defects herein shall constitute initial mold acceptance. Subsequently, the first and last pieces from each mold cavity from each days production (after a minimum mold preheat of one hour) inspected for the applicable classification of defects herein shall constitute lot acceptance. In addition, any time the mold requires repair an additional five pieces from each mold cavity (after a minimum mold preheat of one hour) shall be inspected for the applicable classification of defects herein before resumption of production. If any sample fails to meet any of the applicable inspection requirements, the lot of production under consideration shall be rejected.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" columns in the tables starting with 4.4.2.1. Inspection equipment designs shall be submitted for approval in accordance with the terms of the contract (see Section 6 of MIL-A-48078 and 6.3 herein).

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4.5 Test methods and procedures.

4.5.1 Molding preforms.

4.5.1.1 Chemical composition (see 3.3.1). Determine the chemical composition of the molding compound by oven and muffle furnace gravimetric analysis. Weigh a 5 to 6 gram sample of the molding compound to the nearest tenth of a milligram (0.0001 gm) into a previously fired (650°C) and tared crucible. Condition the sample and crucible for twenty minutes in a circulating air oven conditioned at 148 ± 3°C. Remove, cool in a desiccator, reweigh and further condition in a muffle furnace at 650 ± 3°C for three hours. Cool again in a desiccator and reweigh to obtain the final sample weight. Calculate the percentage composition as follows:

$$V = \frac{W_1 - W_2 \times 100}{W_1}$$

$$R = \frac{W_2 - W_3 \times 100}{W_1}$$

$$G = \frac{W_3 \times 100}{W_1}$$

Where:

V = % volatiles
R = % resin solids
G = % fiberglass
W_1 = Original sample weight
W_2 = Sample weight after 20 minutes at 148 ± 3°C
W_3 = Sample weight after 20 minutes at 148 ± 3°C and 3 hours at 650 ± 3°C.

4.5.1.2 Infrared spectrum of resin extract (see 3.3.2). Put a 5 to 6 gram sample of the molding compound into a beaker and add 50 milliliter (ml) of acetone. Filter the dissolved resin into a 100 ml evaporating dish and allow the acetone to evaporate. Smear a small sample of the residual resin and other solubles onto an infrared transparent window and analyze and record its infrared spectrum with an infrared spectrophotometer.

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4.5.1.3 Curing properties by differential screening calorimetry (See 3.3.3 and 3.4.1). Place a small portion of residual uncured resin and other solubles, obtained by acetone extraction as described in 4.5.1.2, in a hermetic aluminum cup and seal with a hermetic aluminum lid in a sample encapsulating press (employing suitable dies). Record the differential scanning calorimeter scan of the sealed sample VS an empty sealed reference cup using a Du Pont differential scanning calorimetry cell atop a Du Pont 900 differential thermal analyzer or approved equivalents.

4.5.2 Molded insulators.

4.5.2.1 Density (see 3.4.2). The density of complete bulkhead insulators shall be determined in accordance with ASTM D792 (the 50 gm weight restriction shall not apply).

4.5.2.2 Transition temperature (see 3.4.3). A Dupont 941 thermomechanical analyzer in conjunction with a Dupont 900 differential thermal analyzer (X-Y recorder) or approved equivalents shall be used for this test. A 3/8" x 3/8" specimen with flat parallel sides shall be cut from the molded insulator and centrally positioned on the bottom stage of the quartz sample tube holder. The furnace shall then be fastened surrounding the tube holder and specimen. Using the Y axis zero shift, set the Y axis zero near the bottom of the chart paper. Lower an unweighted flat expansion probe until it just touches the sample and the transformer collar is adjusted to bring the Y axis pen to the zero point. To begin the program lower the pen, adjust the heating rate to 5°C/minute and set the program mode dial to "Heat". The program shall be stopped when the temperature of the sample reaches 200°C as shown on a typical plot (see Figure 3). To determine the transition temperature, tangents to the straight line portions of the curve above and below the transition temperature are drawn. The intersection of these tangents shall be taken as the transition temperature (T_g).

4.5.2.3 Flexure (see 3.4.4). Flexure tests shall be performed on complete insulators using approved test equipment and fixtures.

4.5.2.4 Voids and porosity (see 3.4.5). Qualitative inspection for voids and excess porosity shall be performed by cutting the bulkhead insulator approximately in half along any diametric line and examining the exposed surfaces after saturation with water, alcohol or acetone. Voids or excess porosity will be indicated as dark spots, areas or lines as the applied liquid evaporates from the newly cut interior surface (any defect, being more porous than the surrounding material, will retain the applied liquid longer and become evident as a dark area).

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4.5.2.5 Infrared inspection. Infrared inspection shall be performed with approved test equipment, standards, calibration instructions and operating procedures.

5. PACKAGING

5.1 Packaging, packing and marking. Each insulator shall be individually wrapped in chemically neutral wrapping paper meeting the requirements of MIL-P-17667 and secured with pressure sensitive adhesive tape meeting the requirements of FED-SPEC PPP-T-42. The individual insulators shall then be fully overwrapped with at least one layer of cellulosic cushioning material meeting the requirements of FED-SPEC PPP-C-843 and secured with PPP-T-42 pressure sensitive adhesive tape. Five individually wrapped insulators shall then be placed pin-wheel fashion with a sixth in the center in metal shipping drum 9282999-70. Successive layers of six insulators each shall be placed in the drum until a total quantity of seventy-two items has been reached. Additional cellulosic cushioning shall be added over and between layers to provide a tight pack. The cover of the metal shipping drum shall then be installed and sealed in accordance with 8806652. Shipping drums shall be marked in accordance with dwg. 8806652.

6. NOTES

6.1 Intended use. Bulkhead insulators covered by this specification are intended for use on rocket motor/bulkhead assemblies for XM753, 8 Inch Projectiles.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. First article inspection, when required (see 3.5).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD1423 Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be

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delivered by the contractor in accordance with the contact or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph no.	Data requirements	Applicable DID no.
4.4.4	Inspection equipment approval	DI-R-1714 (Tailored)

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Equipment design approval. Submit equipment designs, as required, to Commander U S Army Armament Research and Development Command, ATTN: DRDAR-QAN-I, Dover, New Jersey 07801.

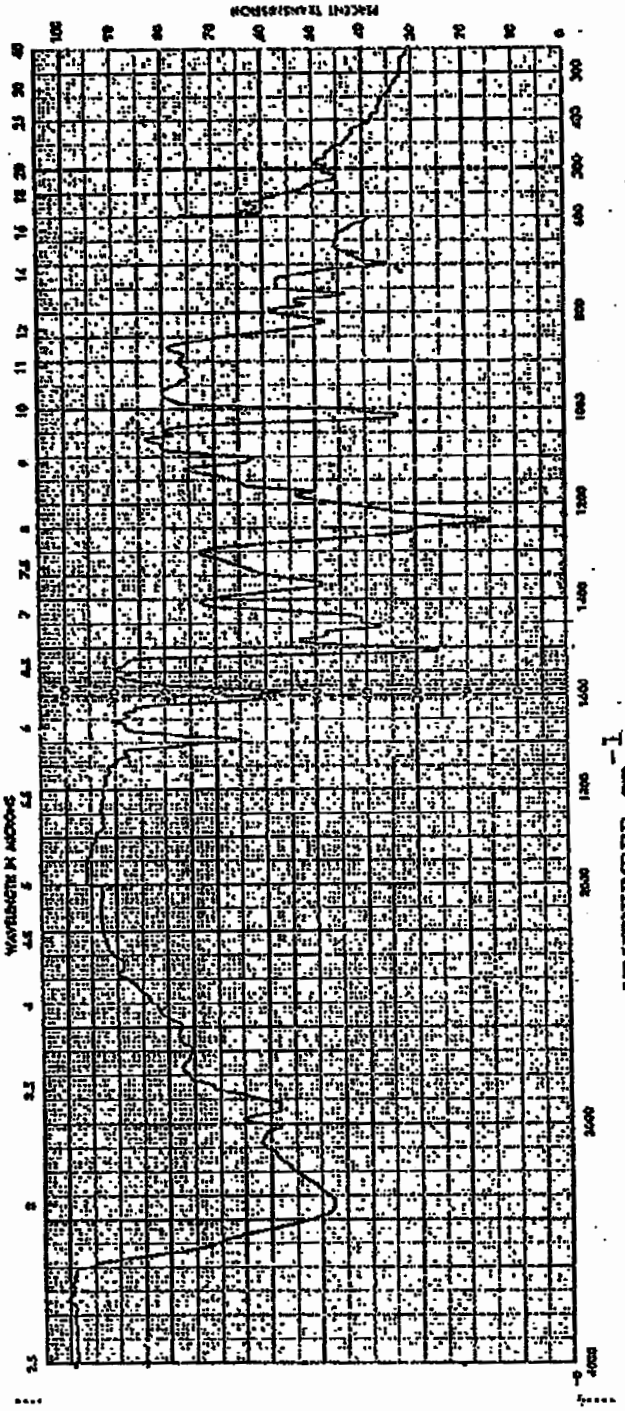
6.4 Molding compound approval. Each lot of molding compound preforms shall be approved by Commander US Army Armament Research and Development Command ATT: DRDAR-LCA-OP, Dover, New Jersey 07801 or an approved laboratory.

6.5. Objective evidence. Documentary evidence that parts and materials comply with applicable dwg. and specification requirements.

6.6 Supersession data. This specification includes the requirements of Purchase Description AR-PD-32 dated 10 January 1980 with Amendment 2 dated 11 June 1980.

6.7 Drawings and specifications. Drawings and specifications listed in Section 2 of this specification under the heading US Army Armament Research and Development Command may also include dwgs. and specifications prepared and identified as Frankford Arsenal, Edgewood Arsenal, Rock Island Arsenal or Picatinny Arsenal dwgs. and specifications. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

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NOTES:
1. Wavelength tolerance ±10 cm^{-1}

FIGURE 1. Spectrum of resin.

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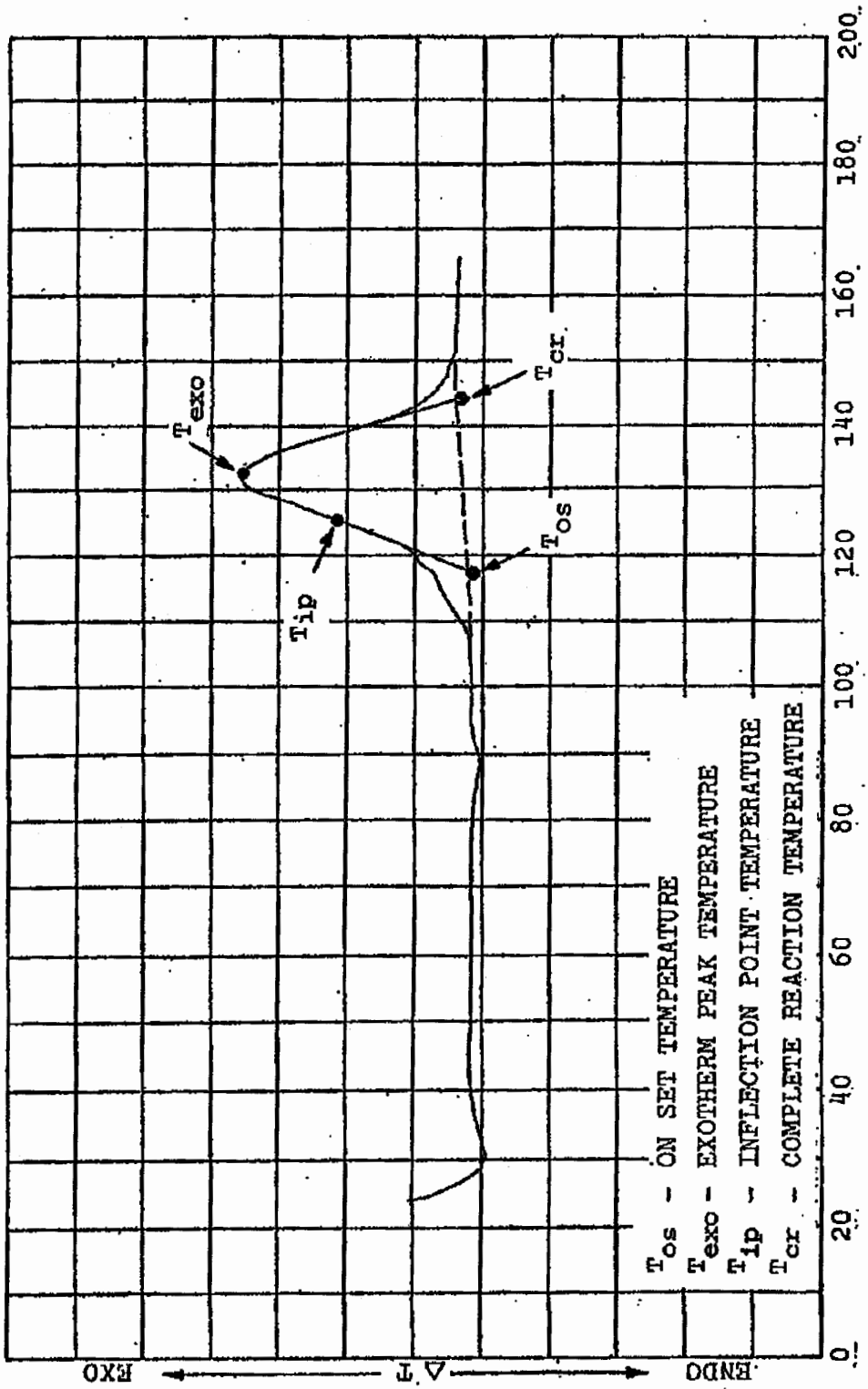


FIGURE 2. Typical plot of curing properties.

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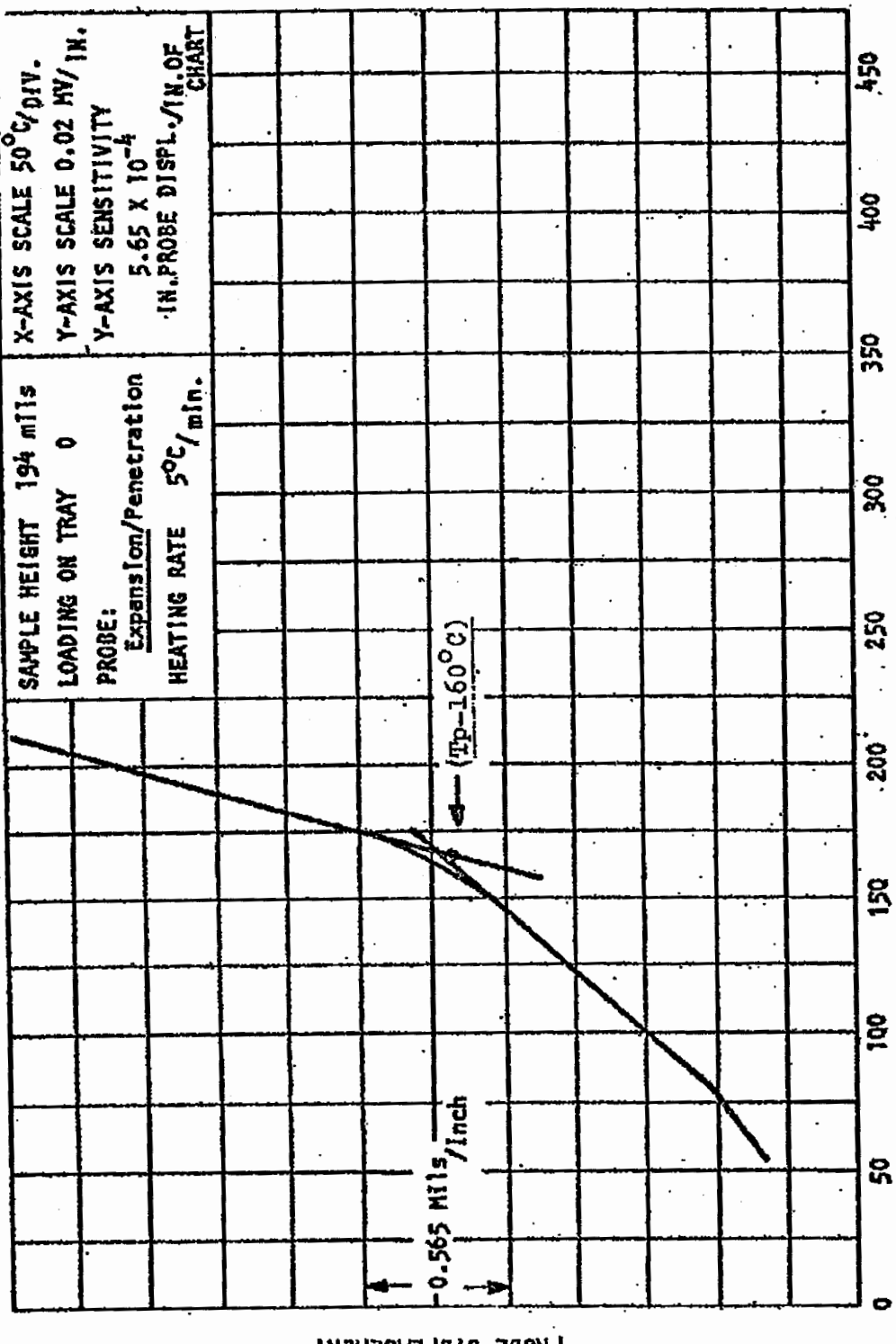


FIGURE 3. Typical plot of transition temperature.

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